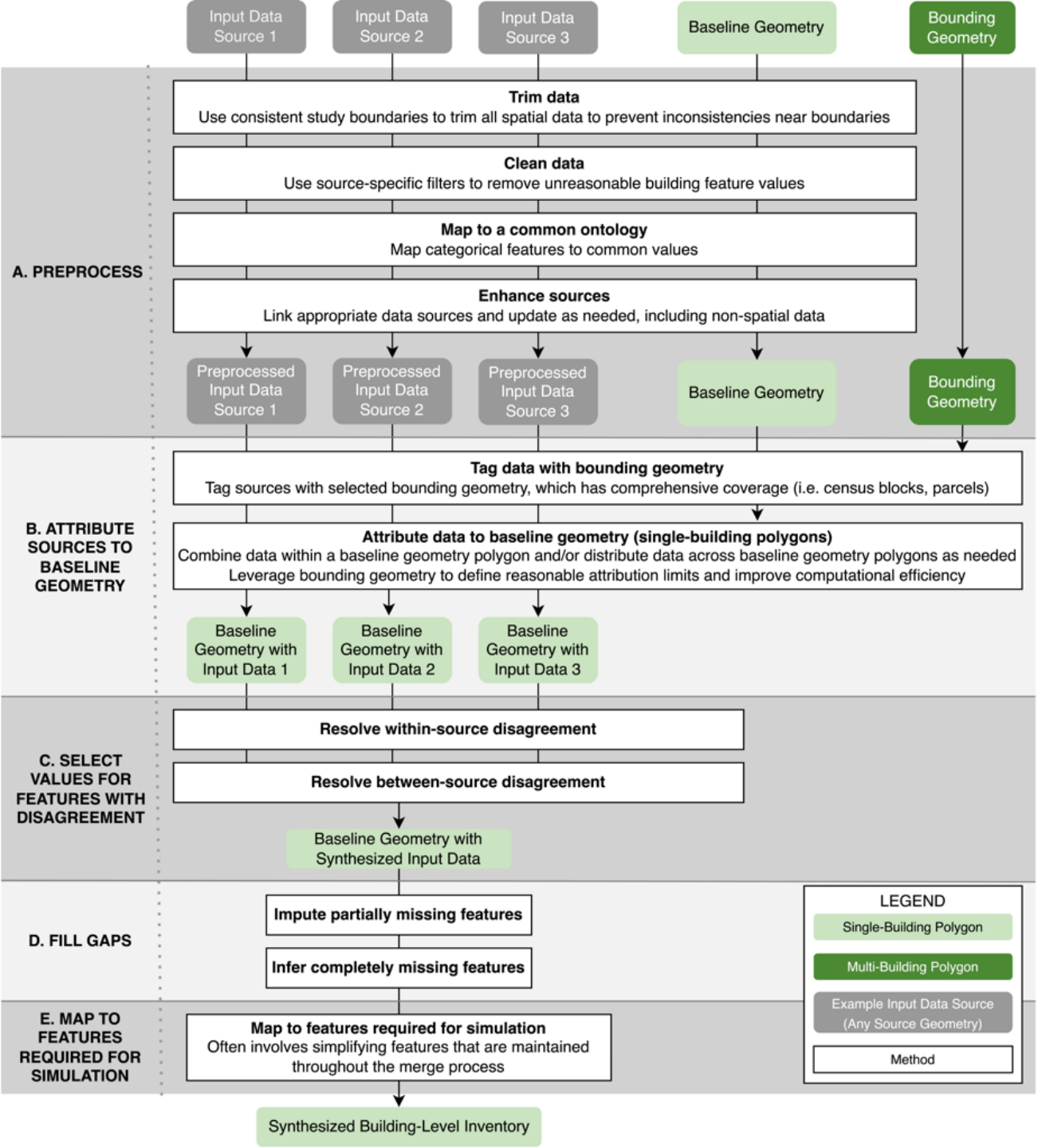
Input Data Source 1
Input Data Source 2
Input Data Source 3
Baseline Geometry
Bounding Geometry
A. PREPROCESS
Trim data
Use consistent study boundaries to trim all spatial data to prevent inconsistencies near boundaries
Clean data
Use source-specific filters to remove unreasonable building feature values
Map to a common ontology
Map categorical features to common values
Enhance sources
Link appropriate data sources and update as needed, including non-spatial data
Preprocessed Input Data Source 1
Preprocessed Input Data Source 2
Preprocessed Input Data Source 3
Baseline Geometry
Bounding Geometry
B. ATTRIBUTE SOURCES TO BASELINE GEOMETRY
Tag data with bounding geometry
Tag sources with selected bounding geometry, which has comprehensive coverage (i.e. census blocks, parcels)
Attribute data to baseline geometry (single-building polygons)
Combine data within a baseline geometry polygon and/or distribute data across baseline geometry polygons as needed
Leverage bounding geometry to define reasonable attribution limits and improve computational efficiency
Baseline Geometry with Input Data 1
Baseline Geometry with Input Data 2
Baseline Geometry with Input Data 3
C. SELECT VALUES FOR FEATURES WITH DISAGREEMENT
Resolve within-source disagreement
Resolve between-source disagreement
Baseline Geometry with Synthesized Input Data
D. FILL GAPS
Impute partially missing features
Infer completely missing features
E. MAP TO FEATURES REQUIRED FOR SIMULATION
Map to features required for simulation
Often involves simplifying features that are maintained throughout the merge process
Synthesized Building-Level Inventory
LEGEND
Single-Building Polygon
Multi-Building Polygon
Example Input Data Source (Any Source Geometry)
Method